



American International University-Bangladesh

Mid-Term Assignment

Course Title: Compiler Design

CourseCode: CSC-3216

Section: E

Submitted on 17-03-2020 by-

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□

Answer to the ques no. 02

Grammar	Semantic rules
$S \rightarrow TL$	$L.in = T.type$
$T \rightarrow int$	$T.type = INT$
$T \rightarrow float$	$T.type = FLOAT$
$T \rightarrow char$	$T.type = CHAR$
$T \rightarrow double$	$T.type = DOUBLE$
$T \rightarrow L, id$	$L.in = L.in$ $addtype(L.in, id.entry)$
$L \rightarrow id$	$addtype(L.in, id.entry)$

□

Answer to the ques no. 03Input statement: $a = b^* - c + b^* - c$

The three address code is:

tem 1 = Uninus c

tem 2 = b^* tem 1

tem 3 = Uninus c

tem 4 = b^* tem 3

$$\text{tem } 5 = \text{tem } 2 + \text{tem } 4$$

$$a = \text{tem } 5.$$



Answer to the ques: 4

B1:

for ($i = 1$ to n)

B2:

$J = 1$
while ($J \leq n$)

B3:

$A = B * (C/D)$
 $J = J + 1;$

Basic Block B:

for ($i = 1$ to n)

$J = 1$
while ($J \leq n$)

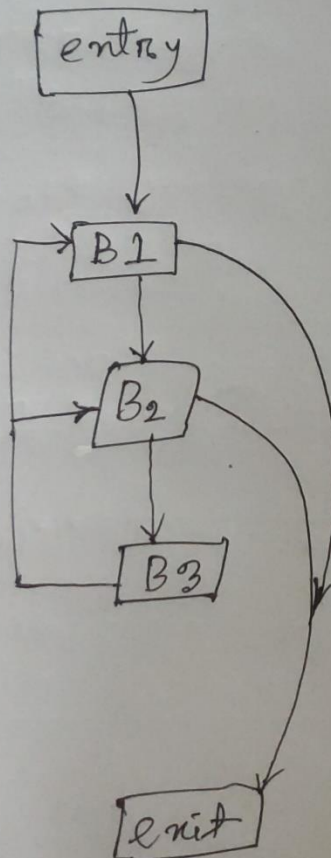
$A = B * C/D$
 $J = J + 1;$

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Data flow Graph



The end.